1-month versus 12-month dual antithrombotic therapy after percutaneous coronary intervention in patients with atrial fibrillation (OPTIMA-AF): a multicentre, open-label, hybrid non-inferiority and superiority, randomised, controlled trial

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Summary

Background **The optimal duration of dual antithrombotic therapy after percutaneous coronary intervention (PCI) in patients with atrial fibrillation remains uncertain. We aimed to compare the efficacy and safety of 1-month versus 12-month dual antithrombotic therapy in this population.**

Methods **OPTIMA-AF was an active-control, randomised, hybrid non-inferiority and superiority trial in which patients with atrial fibrillation undergoing PCI with intravascular imaging guidance were randomly assigned (1:1) to receive 1-month dual antithrombotic therapy (direct oral anticoagulant [DOAC] plus P2Y12 inhibitor) followed by DOAC monotherapy or 12-month dual therapy followed by DOAC monotherapy, across 75 sites in Japan. Eligible patients were aged 20 years or older with non-valvular atrial fibrillation, a $CHADS_2$ score of 1 or higher, undergoing PCI for native coronary lesions for chronic coronary syndrome or unstable angina, with planned post-PCI treatment with a DOAC. Randomisation used web-based central allocation stratified by trial centre. Investigators and patients were unmasked; clinical events were adjudicated by an independent committee masked to treatment. The primary efficacy endpoint was a composite of all-cause death or thromboembolic events at 12 months; the primary safety endpoint was major or clinically relevant non-major bleeding at 12 months. The trial used a fixed-sequence testing strategy of non-inferiority for efficacy, superiority for safety, and superiority for efficacy. Analyses used the full analysis set including all patients who underwent PCI, were randomly assigned to treatment, received at least one dose of study drugs, and met prespecified analysis-set criteria. This trial is registered with the Japan Registry of Clinical Trials (jRCTs051190053), and is complete.**

Findings **From Oct 7, 2019, to Sept 3, 2024, 1101 patients were assessed for eligibility; 1088 were randomly assigned to treatment and 1079 were included in the full analysis set (1-month dual therapy n=542; 12-month dual therapy n=537). Median age was 76 years (IQR 70–81). 225 (21%) of 1079 patients were female and 854 (79%) were male. Median $CHADS_2$ score was 2 (IQR 2–3). Median follow-up was 540 days (IQR 517–559). The primary efficacy endpoint occurred in 29 patients in the 1-month dual therapy group and 23 patients in the 12-month dual therapy group (Kaplan–Meier estimate 5·4% *vs* 4·3%; absolute difference 1·1 percentage points [95% CI –1·5 to 3·6]; hazard ratio [HR] 1·25 [95% CI 0·73–2·17]). The primary safety endpoint occurred in 24 patients versus 47 patients (Kaplan–Meier estimate 4·5% *vs* 8·8%; absolute difference –4·4 percentage points [95% CI –7·3 to –1·4]; HR 0·50 [95% CI 0·30–0·81]; p=0·0041 for superiority).**

Interpretation **Among patients with atrial fibrillation and predominantly chronic coronary syndrome undergoing PCI with intravascular imaging guidance, 1-month dual antithrombotic therapy followed by DOAC monotherapy was non-inferior to 12-month therapy for death or thromboembolic events and reduced major or clinically relevant non-major bleeding at 12 months, suggesting an overall favourable net clinical profile. Efficacy findings should be interpreted with appropriate caution in light of the lower-than-anticipated event rates and fixed absolute non-inferiority margin.**

Funding **Abbott Medical Japan.**

Research in context

Evidence before this study

We searched PubMed and MEDLINE from database inception to Aug 31, 2025, without language restrictions, using terms related to "atrial fibrillation", "percutaneous coronary intervention", "antithrombotic therapy", "oral anticoagulant", "direct oral anticoagulant", "P2Y12 inhibitor", and "randomised controlled trial". We included randomised trials in adults with atrial fibrillation or another indication for oral anticoagulation undergoing percutaneous coronary intervention (PCI) that compared dual antithrombotic therapy with triple antithrombotic therapy and reported bleeding and thromboembolic or ischaemic outcomes within 12 months after PCI. We excluded non-randomised studies, trials terminated prematurely with insufficient statistical power, studies in patients without atrial fibrillation or another indication for oral anticoagulation, and studies not focused on post-PCI antithrombotic treatment. We identified five major randomised trials—WOEST, PIONEER AF-PCI, RE-DUAL PCI, AUGUSTUS, and ENTRUST-AF PCI—that informed current practice.

The randomised trials generally used blinded outcome assessment, although most used an open-label design. Collectively, they showed a consistent reduction in bleeding with dual antithrombotic therapy compared with triple antithrombotic therapy, whereas effects on thromboembolic or ischaemic outcomes remained less certain. Relevant meta-analyses of randomised trials broadly supported the same direction of effect. However, we found no randomised trial that directly evaluated whether dual antithrombotic therapy could be shortened to 1 month after PCI in patients with atrial fibrillation.

Added value of this study

This randomised trial provides direct evidence on the duration of dual antithrombotic therapy after PCI in patients with atrial fibrillation treated with contemporary drug-eluting stents and direct oral anticoagulants. By testing a 1-month dual therapy strategy against a 12-month strategy, OPTIMA-AF helps to address whether dual therapy can be shortened before transition to direct oral anticoagulant monotherapy. The trial met non-inferiority for the primary efficacy endpoint of all-cause death or thromboembolic events and showed superiority for major or clinically relevant non-major bleeding, although efficacy findings require cautious interpretation because event rates were lower than anticipated.

Implications of all the available evidence

Trials including AFIRE and EPIC-CAD have supported oral anticoagulant monotherapy as long-term antithrombotic therapy in patients with atrial fibrillation and stable coronary artery disease, and AQUATIC provided related evidence in patients with chronic coronary syndrome receiving oral anticoagulation. OPTIMA-AF complements this evidence by addressing the early post-PCI period—predominantly in patients with chronic coronary syndrome—and providing evidence relevant to the timing of de-escalation from combination antithrombotic therapy to DOAC monotherapy. Taken together, the available evidence supports cautious consideration of earlier de-escalation of combination antithrombotic therapy after PCI in patients with atrial fibrillation, with careful attention to bleeding and thromboembolic risks. However, because the observed event rates were lower than anticipated under a fixed non-inferiority margin, efficacy estimates warrant cautious interpretation. Further adequately powered trials are needed to clarify applicability in acute coronary syndrome populations and in clinically important subgroups.

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Introduction

Approximately 5–10% of patients undergoing percutaneous coronary intervention (PCI) with drug-eluting stents have coexisting atrial fibrillation, and therefore require both anticoagulation to prevent stroke and antiplatelet therapy to prevent stent thrombosis.[1,2] This combination substantially increases bleeding risk and represents a major clinical challenge. Historically, the standard antithrombotic approach after PCI was triple therapy, comprising an oral anticoagulant—initially, usually a vitamin K antagonist; more recently, for some patients, a direct oral anticoagulant (DOAC)—alongside dual antiplatelet therapy (DAPT; aspirin and a P2Y12 inhibitor).[3] Randomised trials[4–7] in patients with atrial fibrillation undergoing PCI have shown that dual therapy with a DOAC plus a P2Y12 inhibitor reduces bleeding compared with vitamin K antagonist-based triple therapy, without a clear increase in ischaemic events. European and Japanese guidelines therefore recommend restricting triple therapy to the periprocedural or early post-PCI period (up to 2 weeks) in most patients, followed by dual antithrombotic therapy with a DOAC and a P2Y12 inhibitor (typically clopidogrel) during the first 6–12 months after PCI.[4–9] Beyond 6–12 months after PCI, DOAC monotherapy is generally preferred over continued DOAC plus single antiplatelet therapy in patients with atrial fibrillation, a strategy supported by evidence from randomised trials in stable coronary artery disease.[10–12] However, the combination of DOAC and antiplatelet agents has been reported to increase bleeding risk,[10,11,13,14] suggesting that a shorter duration of dual therapy could be preferable, provided protection against thromboembolic and coronary events is not compromised.

Although several trials—mainly in patients without atrial fibrillation—have established the safety and efficacy of a 1-month DAPT strategy compared with prolonged DAPT,[15] evidence specifically supporting a shortened dual antithrombotic therapy regimen (DOAC plus P2Y12 inhibitor) followed by DOAC monotherapy in patients with atrial fibrillation remains scarce. However, emerging data support the feasibility of this approach. Advances in stent technology have substantially reduced the risk of stent thrombosis, and DOACs have been shown to possess antiplatelet properties.[16–18] Furthermore, appropriate management of atrial fibrillation has been established and implemented in daily clinical practice globally.[19–22] These developments support the hypothesis that shortening dual therapy to 1 month after PCI could reduce bleeding risk without compromising protection against thromboembolic or coronary events in this high-risk population. In this trial, we aimed to evaluate the efficacy and safety of 1-month dual therapy followed by DOAC monotherapy, compared with 12-month dual therapy, after PCI in patients with atrial fibrillation.

Methods

Study design and participants

The OPTIMA-AF trial (optimal antiplatelet therapy in combination with direct oral anticoagulants in patients with non-valvular atrial fibrillation undergoing percutaneous coronary intervention with everolimus-eluting stent) was a prospective, investigator-initiated, multicentre, open-label, blinded-endpoint, active-control, randomised, hybrid non-inferiority and superiority trial. The trial rationale and design have been published previously (appendix 2 p 66).[23] Briefly, the primary objective of the OPTIMA-AF trial was to compare the efficacy and safety of 1-month (short) dual antithrombotic therapy (1-month DOAC plus P2Y12 inhibitor followed by DOAC monotherapy) against 12-month (long) dual antithrombotic therapy (12-month DOAC plus P2Y12 inhibitor followed by DOAC monotherapy) in the treatment of patients with atrial fibrillation and ischaemic heart disease undergoing PCI. Details regarding the participating investigators and trial organisation are provided in appendix 2 (pp 4–8). 75 hospitals across Japan participated in this trial, which was registered with Japan Registry of Clinical Trials on Oct 1, 2019 (jRCTs051190053) and is complete. The study was conducted in accordance with the Declaration of Helsinki and the Ethical Guidelines for Clinical Studies issued by the Japanese Ministry of Health, Labour, and Welfare. Patients and the public were not involved in the design, conduct, reporting, or dissemination plans of this research. The trial received approval from the Certified Review Board of the University of Osaka (a certified research ethics committee by the Japanese Clinical Research Act; S18031). Study progress was monitored centrally and, when necessary, on site. An independent data and safety monitoring board reviewed unblinded data summaries annually (Dec 10, 2021; Sept 22, 2022; Oct 2, 2023; Oct 4, 2024; and Oct 2, 2025). The same independent members served as both the clinical event committee and the data and safety monitoring board. The committee was authorised to recommend protocol modifications, temporary suspension, or early termination of the trial if safety concerns arose.

See Online for appendix 2

For trial registration see https://jrct.mhlw.go.jp/en-latest-detail/jRCTs051190053

Patient enrolment occurred between 2019 and 2024, overlapping with the COVID-19 pandemic. Several centres temporarily paused enrolment in 2020 because of institutional restrictions. Follow-up visits scheduled during these periods were conducted by telephone whenever feasible. All endpoint events—including deaths and thromboembolic or bleeding events—were verified through patient interviews and hospital records. No protocol amendments affecting the primary endpoint definition, sample size, or statistical analysis plan were made due to the COVID-19 pandemic. The overall enrolment period was extended by 2 years to ensure target sample completion.

Participants

Patients with non-valvular atrial fibrillation (ie, atrial fibrillation not associated with rheumatic mitral stenosis or prosthetic valve replacement), a $CHADS_2$ score of 1 or higher, and an indication for PCI for native coronary lesions were eligible for this trial. Other key inclusion criteria were: planned continuation of adjunctive oral anticoagulation with a DOAC after PCI; an indication for PCI due to chronic coronary syndrome (stable angina or silent myocardial ischaemia) or acute coronary syndrome (restricted to unstable angina); suitability for PCI based on local practice (operator's judgement or heart team decision); and age of at least 20 years. Key exclusion criteria included ST elevation myocardial infarction and non-ST elevation myocardial infarction, planned staged PCI procedure more than 2 weeks after the index procedure, life expectancy of less than 1 year, and severe renal insufficiency (creatinine clearance <15 mL/min or maintenance dialysis). Full criteria are listed in appendix 2 (p 32).

Patients were recruited consecutively at participating centres, although the final decision for enrolment was at the discretion of the treating physician. Sex data were taken from medical records. Race or ethnicity data were not formally collected; however, because all patients were recruited in Japan, the study population was likely to have been predominantly east Asian. All patients provided written informed consent before enrolment.

Randomisation and masking

Eligible patients who had successfully undergone PCI were randomly assigned in a 1:1 ratio to the 1-month dual therapy group or the 12-month dual therapy group. Randomisation was implemented through a secure, web-based central system using a computer-generated allocation sequence managed by an independent electronic data capture administrator, with stratified simple randomisation by centre. Recruiting investigators enrolled patients and performed registration and randomisation via the central system but had no access to the underlying sequence. Allocation was concealed until registration was completed, at which point the system automatically disclosed assignment. The trial was open-label and patients and treating physicians were aware of treatment allocation; however, clinical events were adjudicated by an independent clinical event committee of three interventional cardiologists, who were masked to treatment allocation and used blinded individual case materials. Two matching votes were required to finalise each classification. The same members also served the data and safety monitoring function, reviewing aggregate unblinded safety summaries by treatment group at prespecified intervals. The clinical events adjudication committee members were independent of the steering committee, investigators, sponsor, and study operations, and had no role in patient recruitment, treatment allocation, or data analysis. The success of masking of outcome adjudicators was not formally evaluated. Further details are provided in appendix 2 (p 9).

Procedures

PCI was performed by interventional cardiologists at the participating centres using the Xience everolimus-eluting stent (Abbott Vascular, Santa Clara, CA, USA). Additional procedural techniques including pre-dilatation, atherectomy or other lesion modification, and post-dilatation were at the operators' discretion. Imaging with optical coherence tomography or intravascular ultrasound was required before and after PCI. Treatment of multiple vessels and staged procedures within 14 days, or up to 28 days when required, were permitted under the same procedural requirements.

All patients were to receive aspirin plus a P2Y12 inhibitor (clopidogrel or prasugrel) before or at the time of PCI. Patients taking warfarin were switched to a DOAC at least 1 week before the procedure. The 1-month dual therapy group received DOAC plus clopidogrel 75 mg daily or prasugrel 3·75 mg daily for 1 month, then DOAC monotherapy; the 12-month dual therapy group received the same dual therapy for 12 months before switching to DOAC alone. Aspirin was to be stopped immediately after PCI, although add-on aspirin could be used for up to 1 month at the physician's discretion. DOACs (dabigatran, rivaroxaban, apixaban, or edoxaban) were prescribed according to standard stroke-prevention criteria, and patients were encouraged to receive other medical therapy in accordance with Japanese Circulation Society guidance for prevention of cardiovascular disease.[24] Details regarding the trial regimens are provided in appendix 2 (p 66).

Clinical follow-up was scheduled at 1, 12, and 18 months after the index PCI (prespecified visit windows: day 30 [–7 to +14 days], and 12 and 18 months [±1 month]). Outcomes were primarily ascertained during on-site visits, but telephone contact or review of referral letters from attending physicians was also acceptable.

Outcomes

The prespecified primary efficacy endpoint was the composite of all-cause death or thromboembolic events (myocardial infarction, definite stent thrombosis, stroke, or systemic embolism), assessed at 12 months. The primary safety endpoint was International Society on Thrombosis and Haemostasis-defined major bleeding or clinically relevant non-major bleeding, also assessed at 12 months.[25,26] Secondary endpoints included the composite of death, thromboembolic events, or bleeding; the individual components of the primary efficacy endpoint; bleeding events defined by Bleeding Academic Research Consortium and Thrombolysis In Myocardial Infarction criteria; and target lesion, target vessel, and all coronary revascularisation procedures. A detailed list and definitions of all prespecified trial outcomes are provided in appendix 2 (pp 33–38). Prespecified adverse events and clinical endpoint events were collected irrespective of causal relationship. Events suspected to be related to the trial were reported in accordance with the requirements of the Japanese Clinical Trials Act to the certified review board and the Japanese Ministry of Health, Labour, and Welfare. Other events not prespecified as trial endpoints and not considered trial-related were not systematically captured.

Statistical analysis

The required sample size was calculated for a non-inferiority test of the primary efficacy endpoint (death or thromboembolic events), using a two-sided z-test with pooled variance (one-sided α=0·05, two-sided α=0·10) and 85% power. Assuming an event rate of 10% in both groups[4] and an absolute non-inferiority margin of 5·0 percentage points, 1036 patients were required. The same sample size provided 99·2% power at a two-sided α=0·10 to detect a reduction in the primary safety endpoint (bleeding) from 25% to 15%.[4,27] Allowing for 5% attrition, the target enrolment was 1090 patients. The 5·0 percentage point non-inferiority margin was selected as half of the assumed 10% control group event rate and reflected the anticipated balance between a possible loss of efficacy and a reduction in bleeding; a detailed rationale is provided in appendix 2 (p 9).

We used a prespecified fixed-sequence (hierarchical) testing procedure to control multiplicity (familywise type I error). The sequence was: non-inferiority for the primary efficacy endpoint; superiority for the primary safety endpoint; and superiority for the primary efficacy endpoint. Testing proceeded to the next step only if the null hypothesis was rejected at the previous step; otherwise, no further hypothesis testing was performed. Analyses used the full analysis set that included all patients who underwent PCI, were randomly assigned, and received at least one dose of the study drugs, and excluded patients who discontinued the trial before hospital discharge and one patient who was lost to follow-up after discharge before 1-month protocol-treatment status could be confirmed. The patient lost to follow-up after discharge was retained in the safety analysis set. These exclusions were applied according to prespecified case review guidelines before data lock, with classifications confirmed during the blinded data review meeting. The primary analysis of the primary efficacy endpoint was based on the between-group risk difference, evaluated using Miettinen and Nurminen's CI.[28] Non-inferiority was concluded if the upper limit of the two-sided 95% CI for the 12-month (day 360) risk difference was 5·0 percentage points or less. The protocol and statistical analysis plan prespecified non-inferiority testing at a one-sided α of 0·05, equivalent to a two-sided 90% CI, and superiority testing at a two-sided α of 0·10 within the fixed-sequence testing procedure. In this Article, two-sided 95% CIs are presented and superiority tests are reported at a two-sided α of 0·05 for consistency across the manuscript. Superiority testing for the primary safety endpoint and the primary efficacy endpoint was performed using Fisher's exact test. The same primary endpoint analyses were repeated as supplementary analyses in the per-protocol analysis set, defined as patients in the full analysis set without major protocol deviations affecting protocol adherence or endpoint assessment (appendix 2 pp 11–31, 39–64).

Cumulative events were estimated by the Kaplan–Meier method, and treatment effects by Cox

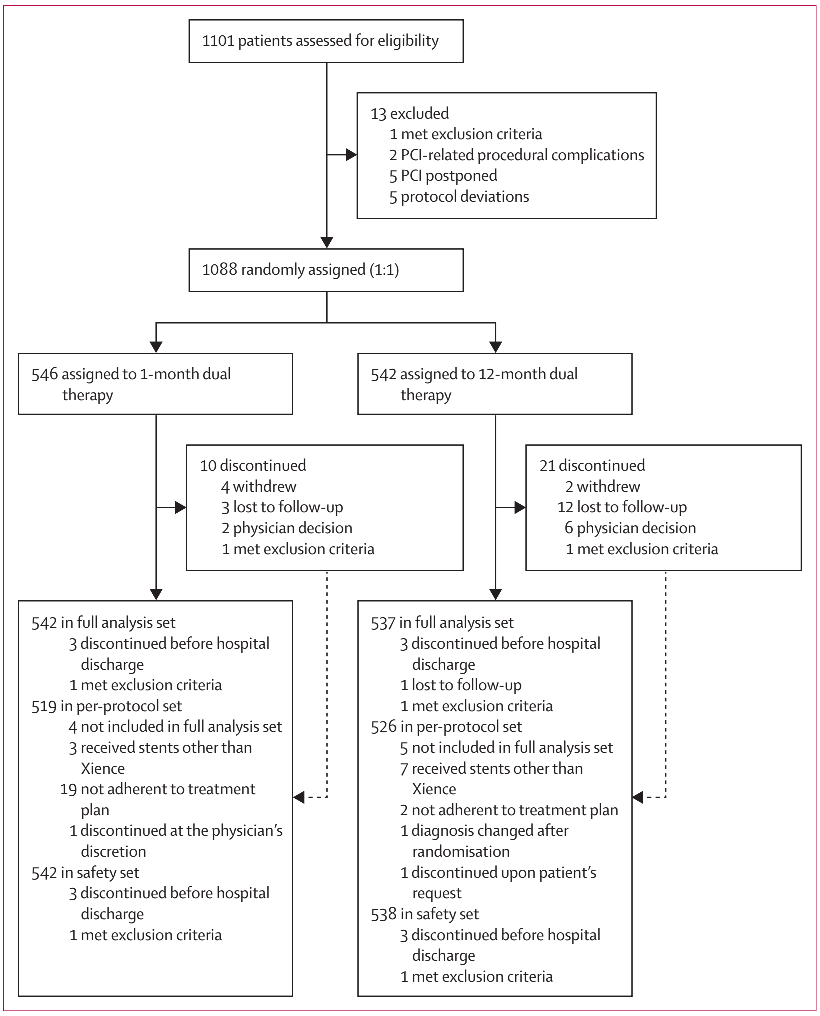


Figure 1: **Trial profile**
The number assessed for eligibility represents electronic data capture pre-registered patients judged by investigators as likely eligible (per inclusion and exclusion criteria), rather than the total number screened at sites. Patients were randomly assigned in a 1:1 ratio to receive 1-month dual antithrombotic therapy with a DOAC plus a P2Y12 inhibitor (clopidogrel or prasugrel), followed by DOAC monotherapy, or 12-month dual antithrombotic therapy with a DOAC plus a single P2Y12 inhibitor, followed by DOAC monotherapy. DOAC=direct oral anticoagulant. PCI=percutaneous coronary intervention.

	1-month dual therapy (n=542)	12-month dual therapy (n=537)
Age, years	76 (71–82)	76 (70–81)
Sex		
Male	438 (81%)	416 (77%)
Female	104 (19%)	121 (23%)
BMI, kg/m²	23·8 (21·6–26·6)	24·2 (22·0–26·4)
Diabetes	208 (38%)	216 (40%)
Hypertension	471 (87%)	459 (85%)
Dyslipidaemia	382 (70%)	392 (73%)
Current smoker	49 (9%)	57 (11%)
Chronic coronary syndrome	510 (94%)	508 (95%)
Acute coronary syndrome*	32 (6%)	29 (5%)
Type of atrial fibrillation		
Paroxysmal	252 (46%)	266 (50%)
Persistent	122 (23%)	103 (19%)
Long-standing persistent	159 (29%)	162 (30%)
Unknown	9 (2%)	6 (1%)
History of cerebral infarction	71 (13%)	61 (11%)
Creatinine clearance, mL/min†	56·4 (43·2–73·5)	57·3 (43·1–74·7)
CHADS$_2$ score	2 (2–3)	2 (2–3)
CHA$_2$DS$_2$-VASc score	4 (3–5)	4 (3–5)
HAS-BLED score	2 (2–3)	2 (2–3)
Type of antiplatelet agents at hospital discharge		
Clopidogrel	263 (49%)	258 (48%)
Prasugrel	273 (50%)	273 (51%)
Aspirin	24 (4%)	29 (5%)
Others	3 (1%)	1 (<1%)
Type of direct oral anticoagulants at hospital discharge		
Dabigatran	23 (4%)	22 (4%)
Rivaroxaban	126 (23%)	119 (22%)
Aspirin Apixaban	142 (26%)	142 (26%)
Edoxaban	251 (46%)	253 (47%)
Not recorded	0	1 (<1%)

Data are median (IQR) or n (%). Percentages might not sum to 100 due to rounding and potential double counting. CHADS$_2$ and CHA$_2$DS$_2$-VASc scores assess stroke risk in patients with atrial fibrillation; HAS-BLED score assesses bleeding risk in patients receiving anticoagulation—higher scores indicate greater risk. *Acute coronary syndrome in this trial refers to unstable angina only (myocardial infarction was excluded). †Creatinine clearance was assessed with the Cockcroft–Gault formula.

Table 1: **Baseline characteristics of the full analysis set**

proportional-hazards regression, expressed as hazard ratios (HRs) with 95% CIs. The proportional-hazards assumption was evaluated by visually assessing Schoenfeld residuals against rank-transformed survival time; no material violations were observed. Survival analysis techniques were used for all time-to-event variables, including the individual components of the primary and secondary endpoints. A 1-month landmark and prespecified subgroup analyses were performed. The protocol and statistical analysis plan prespecified assessment at 12 months. Follow-up was censored at day 360 for the Kaplan–Meier estimates and Cox proportional-hazards models, using the locked analysis dataset. Analyses using the prespecified 12-month window are shown in appendix 2 (p 58), and extended 18-month analyses are shown in appendix 2 (pp 26–31). The day 360 dataset used for these primary analyses was finalised and locked before analysis, with no further accrual of follow-up data or modification of adjudicated event information thereafter. No imputation was applied for missing baseline data; patients lost to follow-up were censored at the time of last contact. No interim analyses or formal stopping guidelines were prespecified. Continuous monitoring of data quality and safety was conducted by the trial data centre throughout the study. Analyses were conducted using SAS version 9.4 (SAS Institute Inc, Cary, NC, USA). Further details are in appendix 2 (p 208).

	1-month dual therapy (n=542)		12-month dual therapy (n=537)		Absolute risk difference	Hazard ratio
	Events	Kaplan–Meier	Events	Kaplan–Meier		
Primary outcomes						
All-cause death or thromboembolic events	29 (5%)	5·4 (3·8–7·6)	23 (4%)	4·3 (2·9–6·4)	1·1 (–1·5 to 3·6)	1·25 (0·73–2·17)
Major bleeding or clinically relevant non-major bleeding	24 (4%)	4·5 (3·0–6·6)	47 (9%)	8·8 (6·7–11·6)	–4·4 (–7·3 to –1·4)	0·50 (0·30–0·81)
Secondary outcomes						
All-cause death	22 (4%)	4·1 (2·7–6·1)	15 (3%)	2·8 (1·7–4·6)	1·3 (–0·9 to 3·4)	1·46 (0·76–2·81)
Cardiovascular death	8 (1%)	1·5 (0·7–3·0)	7 (1%)	1·3 (0·6–2·8)	0·2 (–1·2 to 1·6)	1·13 (0·41–3·12)
Non-cardiovascular death	14 (3%)	2·6 (1·6–4·4)	8 (1%)	1·5 (0·8–3·0)	1·1 (–0·6 to 2·8)	1·74 (0·73–4·14)
Myocardial infarction	4 (1%)	0·7 (0·3–2·0)	4 (1%)	0·8 (0·3–2·0)	0·0 (–1·0 to 1·0)	0·99 (0·25–3·96)
Periprocedural myocardial infarction	1 (<1%)	0·2 (0·0–1·3)	2 (<1%)	0·4 (0·1–1·5)	–0·2 (–0·8 to 0·4)	0·50 (0·04–5·46)
Spontaneous myocardial infarction	3 (1%)	0·6 (0·2–1·7)	2 (<1%)	0·4 (0·1–1·5)	0·2 (–0·6 to 1·0)	1·49 (0·25–8·90)
Stent thrombosis (ARC definition)						
Any stent thrombosis	7 (1%)	1·3 (0·6–2·7)	2 (<1%)	0·4 (0·1–1·5)	0·9 (–0·2 to 2·0)	3·46 (0·72–16·65)
Definite stent thrombosis	1 (<1%)	0·2 (0·0–1·3)	1 (<1%)	0·2 (0·0–1·3)	0·0 (–0·5 to 0·5)	0·99 (0·06–15·78)
Probable stent thrombosis	1 (<1%)	0·2 (0·0–1·3)	1 (<1%)	0·2 (0·0–1·3)	0·0 (–0·5 to 0·5)	0·99 (0·06–15·80)
Possible stent thrombosis	5 (1%)	0·9 (0·4–2·2)	0	0·0 (0·0–0·7)	0·9 (0·1 to 1·8)	Not estimable
Stent thrombosis (ARC-2 definition)						
Any stent thrombosis	1 (<1%)	0·2 (0·0–1·3)	1 (<1%)	0·2 (0·0–1·3)	0·0 (–0·5 to 0·5)	0·99 (0·06–15·78)
Definite stent thrombosis	1 (<1%)	0·2 (0·0–1·3)	1 (<1%)	0·2 (0·0–1·3)	0·0 (–0·5 to 0·5)	0·99 (0·06–15·78)
Probable stent thrombosis	0	0·0 (0·0–0·7)	0	0·0 (0·0–0·7)	0·0 (–0·7 to 0·7)	Not estimable
Silent occlusion	0	0·0 (0·0–0·7)	0	0·0 (0·0–0·7)	0·0 (–0·7 to 0·7)	Not estimable
Stroke	4 (1%)	0·7 (0·3–2·0)	4* (1%)	0·8 (0·3–2·0)	0·0 (–1·0 to 1·0)	0·99 (0·25–3·96)
Ischaemic (cerebral thrombosis or embolism)	3 (1%)	0·6 (0·2–1·7)	4 (1%)	0·8 (0·3–2·0)	–0·2 (–1·2 to 0·8)	0·74 (0·17–3·32)
Haemorrhagic stroke (intracerebral haemorrhage)	1 (<1%)	0·2 (0·0–1·3)	1 (<1%)	0·2 (0·0–1·3)	0·0 (–0·5 to 0·5)	0·99 (0·06–15·79)
Systemic embolism	0	0·0 (0·0–0·7)	0	0·0 (0·0–0·7)	0·0 (–0·7 to 0·7)	Not estimable
Bleeding events†						
Major bleeding	18 (3%)	3·4 (2·1–5·3)	23 (4%)	4·3 (2·9–6·4)	–1·0 (–3·3 to 1·3)	0·77 (0·42–1·43)
Clinically relevant non-major bleeding	7 (1%)	1·3 (0·6–2·7)	26 (5%)	4·9 (3·4–7·1)	–3·6 (–5·7 to –1·5)	0·26 (0·11–0·61)

Data are n (%) or estimates (95% CI). Kaplan–Meier estimates are percentages and absolute risk differences are percentage points. Events are reported through day 360, and Kaplan–Meier estimates are at day 360. Hazard ratios were estimated using Cox proportional-hazards models with follow-up censored at day 360. Hazard ratios were not estimable when one or both treatment groups had no events. Confidence intervals for secondary outcomes have not been adjusted for multiplicity; results should be interpreted with caution. Bleeding events were assessed primarily according to the International Society on Thrombosis and Haemostasis definition. Bleeding was also classified according to the Bleeding Academic Research Consortium criteria and the Thrombolysis in Myocardial Infarction definition (appendix 2 pp 33–38). ARC=Academic Research Consortium. ARC-2=Academic Research Consortium-2. *One patient had both ischaemic and haemorrhagic stroke. †Component rows count patients with each specific event type and are not mutually exclusive because some patients had more than one type of bleeding event; therefore, the sum of the components exceeds the total composite bleeding count.

***Table 2:* Primary and key secondary endpoints at 12 months after percutaneous coronary intervention in the full analysis set**

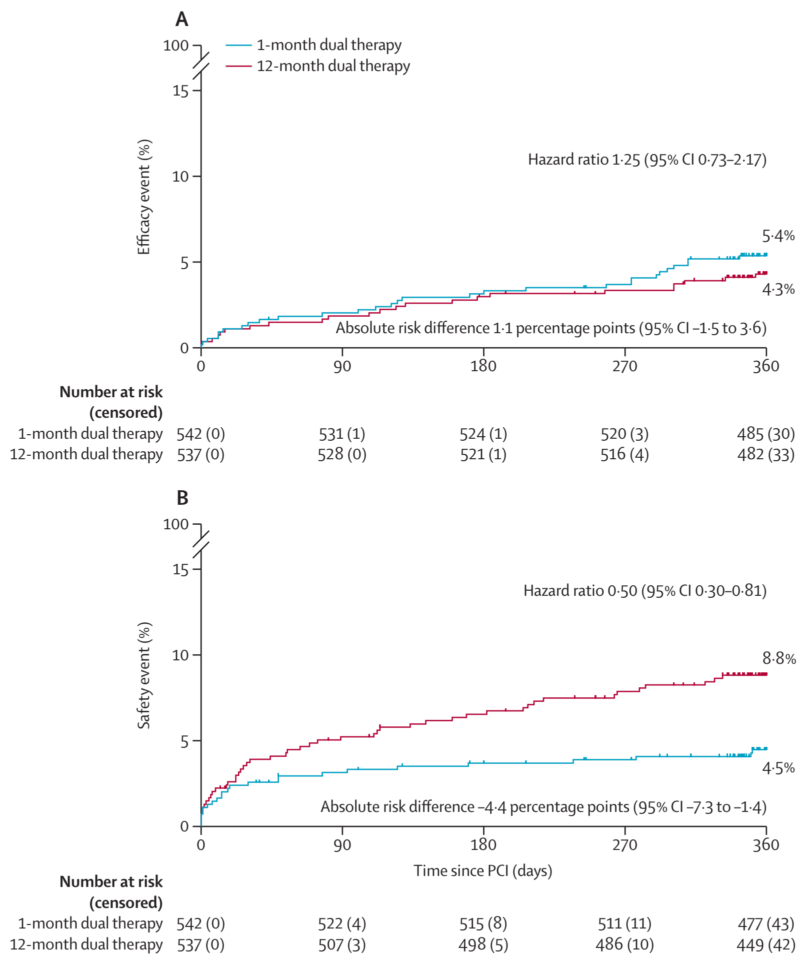


***Figure 2:* Kaplan–Meier analysis of the primary efficacy and safety endpoints (full analysis set)**
(A) The incidence of the primary efficacy endpoint, defined as a composite of all-cause death or thromboembolic events (myocardial infarction, definite stent thrombosis, stroke, or systemic embolism). (B) The primary safety endpoint, defined as bleeding (International Society on Thrombosis and Haemostasis major or clinically relevant non-major bleeding). Percentages are Kaplan–Meier estimates at 12 months (day 360). Hazard ratios compare the 1-month dual therapy group with the 12-month dual therapy group and were estimated using Cox proportional-hazards models with follow-up censored at day 360. PCI=percutaneous coronary intervention.

	1-month dual therapy (n=542)	12-month dual therapy (n=538)
ISTH bleeding classification		
Major bleeding	18/542 (3%)	24/538 (4%)
Clinically relevant non-major bleeding	7/542 (1%)	26/538 (5%)
BARC bleeding classification		
BARC type 2–5	23/542 (4%)	48/538 (9%)
BARC type 3–5	14/542 (3%)	22/538 (4%)

Data are n/N (%) with events to day 360 in the safety analysis set. Detailed endpoint definitions are provided in appendix 2 (pp 34–38). BARC=Bleeding Academic Research Consortium. ISTH=International Society on Thrombosis and Haemostasis.

***Table 3:* Harms in the safety analysis set**

Role of the funding source
This investigator-initiated study was funded by Abbott Medical Japan. The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

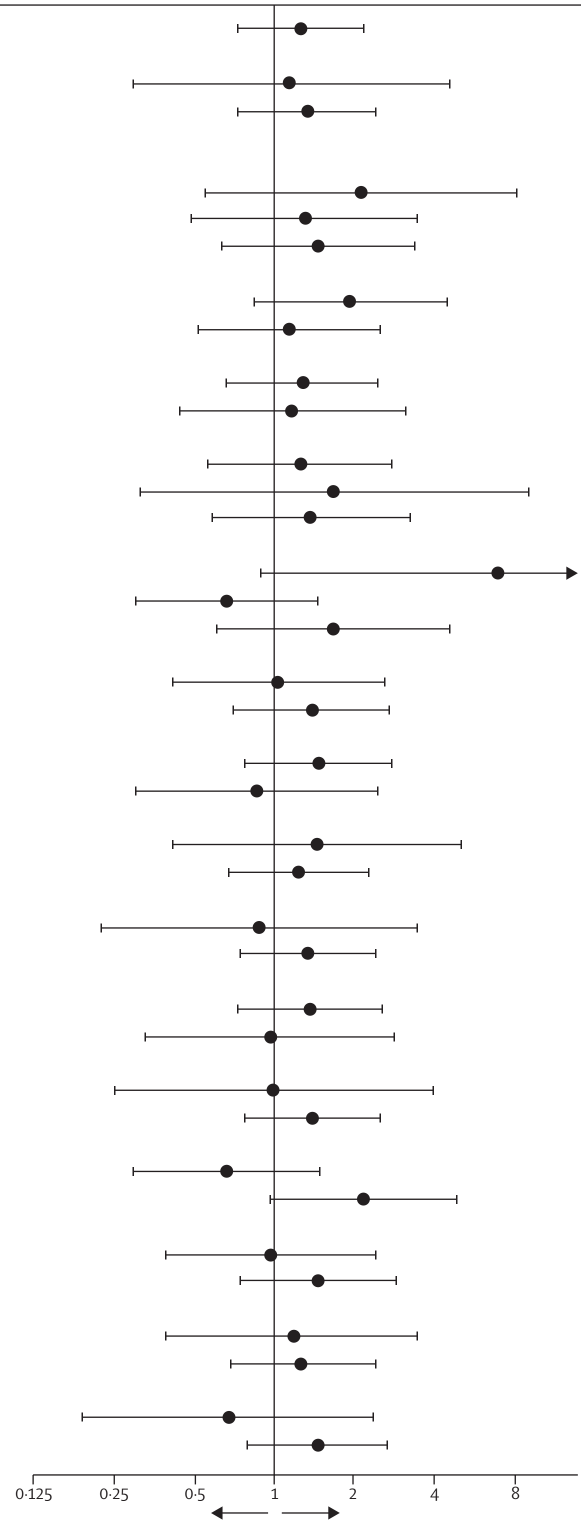
Results
From Oct 7, 2019, to Sept 3, 2024, 1101 patients were assessed for eligibility at 75 sites in Japan. Of these, 1088 were randomly assigned to treatment and 1079 were included in the full analysis set: 542 in the 1-month dual therapy group and 537 in the 12-month dual therapy group (figure 1). Median follow-up was 540 days (IQR 517–559). Baseline characteristics are shown in table 1. The median age was 76 years (IQR 70–81), and 225 (21%) of the 1079 patients were women. Race and ethnicity data were not collected. 518 (48%) patients had paroxysmal atrial fibrillation, 225 (21%) had persistent atrial fibrillation, and 321 (30%) had long-standing persistent atrial fibrillation. The median $CHADS_2$ score was 2 (IQR 2–3), and the median HAS-BLED score was 2 (IQR 2–3). Additional baseline characteristics are shown in appendix 2 (pp 39–42).

PCI was performed mainly via radial access (961 [89%] of 1079 patients). Intravascular imaging guidance was used for 1076 patients (>99%); optical coherence tomography or optical frequency domain imaging for 225 (21%); and intravascular ultrasound for 884 (82%). Complex PCI was performed in 251 patients (23%).[29] A Xience everolimus-eluting stent was implanted in 1073 patients (99%). At hospital discharge, 521 (48%) received clopidogrel and 546 (51%) prasugrel. Dabigatran, rivaroxaban, apixaban, and edoxaban were prescribed in 45 (4%), 245 (23%), 284 (26%), and 504 (47%) patients, respectively, and prescription was not recorded for one patient. The median switch time to DOAC monotherapy was 33 days (IQR 30–39) in the 1-month dual therapy group and 366 days (IQR 347–382) in the 12-month group (appendix 2 p 10). Other details regarding PCI and medication use are provided in appendix 2 (pp 43–48).

In line with the hybrid testing strategy, the 1-month dual therapy met non-inferiority for efficacy and achieved superiority for bleeding, whereas superiority for the efficacy endpoint was not met. In the full analysis set, at 12 months after PCI, the primary efficacy endpoint (a composite of all-cause death or thromboembolic events) occurred in 29 (5%) of 542 patients in the 1-month group and in 23 (4%) of 537 patients in the 12-month group (absolute difference 1·1 percentage points [95% CI –1·5 to 3·6]; HR 1·25 [95% CI 0·73–2·17]; table 2, figure 2A). The primary safety endpoint (major bleeding or clinically relevant non-major bleeding) occurred in 24 (4%) of 542 patients in the 1-month group and in 47 (9%) of 537 patients in the 12-month group (absolute difference –4·4 percentage points [95% CI –7·3 to –1·4]; HR 0·50 [95% CI 0·30–0·81]; p=0·0041

A

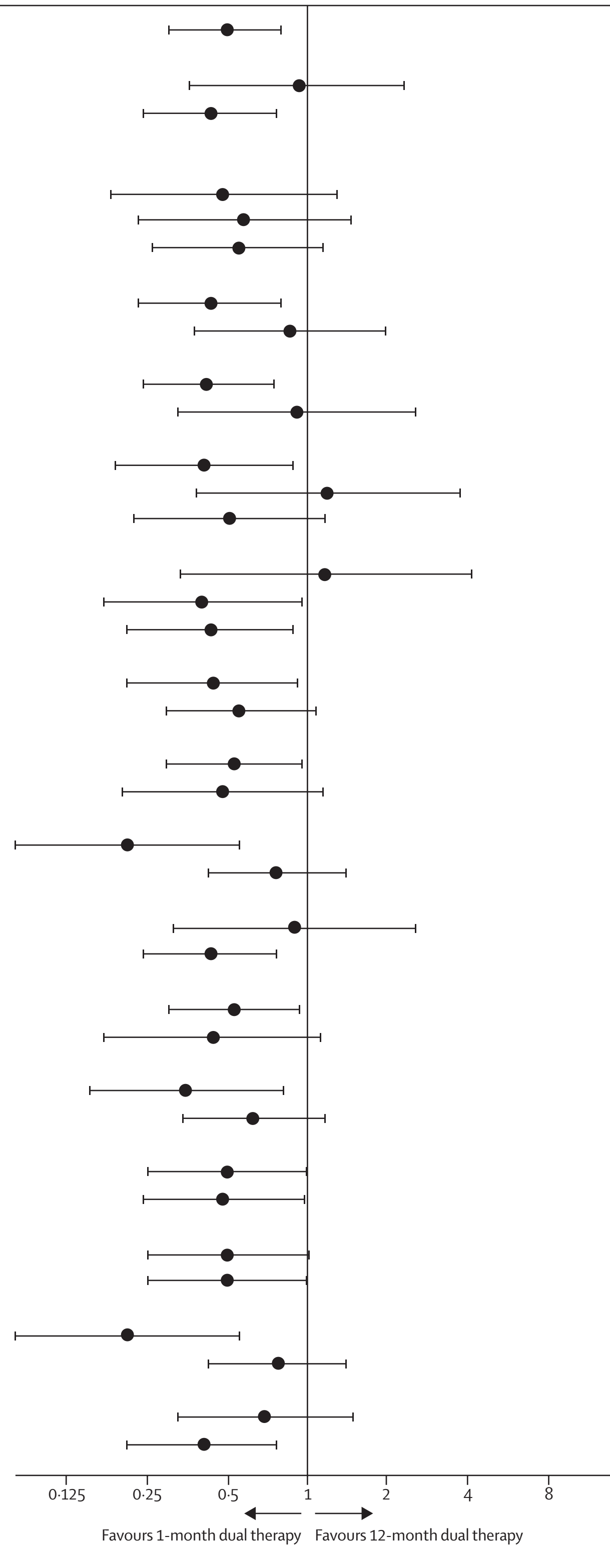
	Number of events/number of patients		Hazard ratio (95% CI)	$p_{interaction}$
	1-month dual therapy	12-month dual therapy		
All patients	**29/542**	**23/537**	**1·25 (0·73–2·17)**	..
Imaging analysis				0·84
OCT	4/100	4/114	1·14 (0·29–4·57)	
Intravascular ultrasound	25/437	18/415	1·33 (0·72–2·43)	
DOAC				0·95
Dabigatran	0/23	3/26	NE	
Rivaroxaban	7/130	3/116	2·12 (0·55–8·21)	
Apixaban	9/142	7/142	1·30 (0·48–3·48)	
Edoxaban	13/247	9/250	1·46 (0·63–3·43)	
DOAC, dose				0·38
Standard	16/374	8/358	1·93 (0·83–4·51)	
Reduced	13/157	12/164	1·15 (0·52–2·52)	
PPI				0·89
Yes	20/409	16/416	1·28 (0·66–2·46)	
No	9/133	7/121	1·17 (0·44–3·15)	
Atrial fibrillation				0·95
Paroxysmal	13/252	11/266	1·25 (0·56–2·79)	
Persistent	4/122	2/103	1·68 (0·31–9·17)	
Long-standing persistent	12/159	9/162	1·37 (0·58–3·25)	
Creatinine clearance, mL/min				0·068
<30	9/40	1/29	7·04 (0·89–55·62)	
30 to <50	10/155	16/164	0·66 (0·30–1·45)	
≥50	10/342	6/342	1·67 (0·61–4·58)	
Diabetes				0·63
Yes	9/208	9/216	1·04 (0·41–2·62)	
No	20/334	14/321	1·38 (0·70–2·73)	
Sex				0·40
Male	23/438	15/416	1·47 (0·77–2·82)	
Female	6/104	8/121	0·86 (0·30–2·49)	
Age				0·82
<75 years	6/235	4/227	1·45 (0·41–5·13)	
≥75 years	23/307	19/310	1·23 (0·67–2·26)	
Vascular size				0·58
<2·5 mm	4/81	4/71	0·88 (0·22–3·50)	
≥2·5 mm	25/458	19/463	1·34 (0·74–2·43)	
DAPT score				0·58
<2	23/416	16/395	1·37 (0·73–2·60)	
≥2	6/126	7/142	0·96 (0·32–2·86)	
PRECISE DAPT score				0·67
<25	4/232	4/235	1·00 (0·25–4·01)	
≥25	25/289	18/285	1·39 (0·76–2·55)	
HAS-BLED score				0·044
<3	9/296	15/328	0·66 (0·29–1·50)	
≥3	20/246	8/209	2·17 (0·96–4·93)	
ORBIT score				0·48
<4	9/345	9/336	0·97 (0·39–2·45)	
≥4	20/197	14/201	1·47 (0·74–2·92)	
CHA_2DS_2-VASc score				0·90
≤3	7/257	6/259	1·18 (0·39–3·50)	
>3	22/285	17/278	1·27 (0·68–2·40)	
Complex PCI				0·27
Yes	4/125	6/126	0·67 (0·19–2·36)	
No	25/417	17/411	1·46 (0·79–2·71)	



(Figure 3 continues on next page)

B

	Number of events/number of patients: 1-month dual therapy	Number of events/number of patients: 12-month dual therapy	Hazard ratio (95% CI)	p interaction
All patients	**24/542**	**47/537**	**0·50 (0·30–0·81)**	..
Imaging analysis				0·17
OCT	8/100	10/114	0·92 (0·36–2·33)	
Intravascular ultrasound	16/437	35/415	0·43 (0·24–0·77)	
DOAC				1·00
Dabigatran	0/23	2/26	NE	
Rivaroxaban	6/130	11/116	0·48 (0·18–1·30)	
Apixaban	7/142	12/142	0·58 (0·23–1·47)	
Edoxaban	11/247	20/250	0·55 (0·26–1·15)	
DOAC, dose				0·18
Standard	14/374	31/358	0·43 (0·23–0·80)	
Reduced	10/157	12/164	0·86 (0·37–2·00)	
PPI				0·21
Yes	17/409	40/416	0·42 (0·24–0·75)	
No	7/133	7/121	0·91 (0·32–2·59)	
Atrial fibrillation				0·30
Paroxysmal	9/252	23/266	0·41 (0·19–0·88)	
Persistent	7/122	5/103	1·19 (0·38–3·74)	
Long-standing persistent	8/159	16/162	0·51 (0·22–1·18)	
Creatinine clearance, mL/min				0·34
<30	6/40	4/29	1·17 (0·33–4·15)	
30 to <50	7/155	18/164	0·40 (0·17–0·96)	
≥50	11/342	25/342	0·43 (0·21–0·88)	
Diabetes				0·66
Yes	10/208	23/216	0·44 (0·21–0·93)	
No	14/334	24/321	0·55 (0·29–1·07)	
Sex				0·85
Male	17/438	30/416	0·53 (0·29–0·96)	
Female	7/104	17/121	0·48 (0·20–1·15)	
Age				0·026
<75 years	5/235	22/227	0·21 (0·08–0·56)	
≥75 years	19/307	25/310	0·77 (0·42–1·40)	
Vascular size				0·23
<2·5 mm	7/81	7/71	0·90 (0·31–2·56)	
≥2·5 mm	17/458	39/463	0·43 (0·24–0·76)	
DAPT score				0·74
<2	18/416	32/395	0·53 (0·30–0·94)	
≥2	6/126	15/142	0·44 (0·17–1·13)	
PRECISE DAPT score				0·27
<25	7/232	20/235	0·35 (0·15–0·82)	
≥25	16/289	25/285	0·63 (0·34–1·18)	
HAS-BLED score				0·94
<3	12/296	26/328	0·50 (0·25–1·00)	
≥3	12/246	21/209	0·48 (0·24–0·98)	
ORBIT score				1·00
<4	12/345	23/336	0·50 (0·25–1·01)	
≥4	12/197	24/201	0·50 (0·25–1·00)	
CHA_2DS_2-VASc score				0·025
≤3	5/257	23/259	0·21 (0·08–0·56)	
>3	19/285	24/278	0·78 (0·42–1·42)	
Complex PCI				0·30
Yes	11/125	16/126	0·69 (0·32–1·49)	
No	13/417	31/411	0·41 (0·21–0·77)	



for superiority; table 2, figure 2B). Only five patients (<1%) in the 1-month group and seven patients (1%) in the 12-month group had both the primary efficacy endpoint and the primary safety endpoint. Sensitivity analyses—including inverse-probability-of-censoring weighting, per-protocol analyses, and analyses accounting for competing risk—were consistent with the main results (appendix 2 p 58). A Kaplan–Meier landmark analysis at 1 month for the primary endpoints and extended follow-up sensitivity analyses through 18 months are provided in appendix 2 (pp 12–13, 26–27).

The incidence of individual components of the primary endpoint and other secondary endpoints are provided in table 2. All-cause mortality was numerically higher in the 1-month group than the 12-month group (4·1% *vs* 2·8%, absolute difference 1·3% [95% CI –0·9 to 3·4]), which was largely reflected by numerically more non-cardiovascular deaths (2·6% *vs* 1·5%, absolute difference 1·1 percentage points [95% CI –0·6 to 2·8]). The incidence of major bleeding was similar between groups (HR 0·77 [95% CI 0·42–1·43]), whereas clinically relevant non-major bleeding was lower with 1-month dual therapy (0·26 [0·11–0·61]). Harms in the safety analysis set, cause of death, and bleeding sites are detailed in table 3 and appendix 2 (pp 62–64).

The effect of 1-month dual therapy on the primary efficacy endpoint was generally consistent across prespecified subgroups, including age, sex, renal function, thrombotic and bleeding risk scores, DOAC type, atrial fibrillation type, and complex PCI, except for HAS-BLED score (figure 3A). Findings for the primary safety endpoint were broadly similar across most prespecified subgroups, with possible heterogeneity according to age and CHA_2DS_2-VASc score (exploratory analysis; figure 3B).

Discussion

In this prospective, multicentre, open-label, active-control, randomised, hybrid non-inferiority and superiority trial, we evaluated the safety and efficacy of 1-month (short) versus 12-month (long) dual antithrombotic therapy in patients with atrial fibrillation undergoing PCI with contemporary drug-eluting stents. The study population consisted predominantly of patients with chronic coronary syndrome (approximately 94%). The trial met its first two primary objectives: non-inferiority of the 1-month short dual therapy for all-cause death or thromboembolic events, and superiority for major or clinically relevant non-major bleeding. Superiority for the composite of all-cause death or thromboembolic events was not met.

To our knowledge, this is the first trial to evaluate whether dual antithrombotic therapy after PCI in patients with atrial fibrillation can be shortened to 1 month before transition to DOAC monotherapy. Previous major trials examined dual therapy durations of 6–12 months, whereas OPTIMA-AF tested a 1-month dual therapy strategy followed by DOAC monotherapy. Unlike earlier studies primarily focused on bleeding endpoints, OPTIMA-AF assessed non-inferiority of the primary efficacy endpoint of all-cause death or thromboembolic events, followed by superiority testing of the primary safety endpoint within a prespecified fixed-sequence procedure. The findings suggest that a 1-month dual regimen followed by DOAC monotherapy reduced major or clinically relevant non-major bleeding, with no evidence that the prespecified non-inferiority margin for the primary efficacy endpoint was exceeded.

Bleeding events occurred less frequently than expected, on the basis of earlier trials, despite the older age of the study population (median 76 years) and universal use of DOACs, both of which are relevant to bleeding risk.[4–7,30] Potential contributors include the shorter duration of dual therapy in the 1-month group; high use of proton pump inhibitors, which was substantially higher in OPTIMA-AF (76·5%) than in two comparable studies and might have contributed to lower gastrointestinal bleeding risk;[10,11] and contemporary procedural and medical management.

The incidence of primary efficacy endpoint events was also lower than anticipated. Several factors could have contributed to this finding. First, approximately 94% of patients had chronic coronary syndrome, with only a small proportion presenting with acute coronary syndrome, whereas earlier trials included a higher proportion of patients with acute coronary syndrome. Second, high statin use (83% in the 1-month group and 86% in the 12-month group; appendix 2 p 47) might have contributed to secondary prevention. Third, nearly all patients underwent PCI with intravascular imaging guidance, which might have contributed to procedural optimisation. These factors might have contributed to the lower-than-anticipated incidence of primary efficacy endpoint events. Although encouraging, the control group incidence of primary efficacy endpoint events was approximately 4%, about half of the 10% anticipated during sample size estimation, highlighting the limitations of a fixed 5 percentage points absolute

Figure 3: **Subgroup analyses of the primary efficacy and safety endpoints (full analysis set)**
Shown are hazard ratios with 95% CIs (1-month dual antithrombotic therapy versus 12-month dual antithrombotic therapy) for the primary efficacy endpoint (A) and the primary safety endpoint (B) according to subgroups in the full analysis set. All subgroups were prespecified in the statistical analysis plan except for the CHA_2DS_2VASc score subgroup, which was conducted as an exploratory analysis. Hazard ratios were estimated using Cox proportional-hazards models with follow-up censored at day 360. p values for interaction were obtained by including a treatment-by-subgroup interaction term in the Cox model. The subgroup analyses for DOAC types are based on data on admission. No multiplicity adjustment was applied. Creatinine clearance was assessed with the Cockcroft–Gault formula. DAPT=dual antiplatelet therapy. DOAC=direct oral anticoagulant. OCT=optical coherence tomography. NE=not estimable. ORBIT=Outcomes Registry for Better Informed Treatment score. PCI=percutaneous coronary intervention. PPI=proton pump inhibitor.

non-inferiority margin under lower-than-expected event rates. The prespecified absolute non-inferiority margin of 5 percentage points was intended to reflect a clinically acceptable trade-off between a possible loss of efficacy and an expected reduction in bleeding. In the observed data, the 12-month risk difference for efficacy was 1·1 percentage points (95% CI –1·5 to 3·6), and major or clinically relevant non-major bleeding decreased by 4·4 percentage points more in the 1-month group than in the 12-month group (4·5% *vs* 8·8%). However, this interval does not exclude an absolute increase of up to 3·6 percentage points in the primary efficacy endpoint, which could be clinically relevant; therefore, efficacy should not be overinterpreted and warrants confirmation in adequately powered trials.

In the prespecified subgroup analyses presented in figure 3, the treatment effect of short dual antithrombotic therapy was consistent across most clinical subsets, including DOAC type and complex PCI, with findings generally consistent with the overall reduction in bleeding and no clear evidence of treatment-effect heterogeneity for the primary efficacy endpoint. Although patients with creatinine clearance lower than 30 mL/min had numerically fewer primary efficacy endpoint events with 12-month dual therapy, the number of patients in this subgroup was small and the estimate was imprecise. Given the high competing bleeding and ischaemic risks in patients with severe chronic kidney disease, decisions on shortening dual therapy in this subgroup should be individualised. Overall, the subgroup findings were generally consistent with the main results, but because these analyses were not powered to establish treatment-effect heterogeneity, apparent differences in treatment effect by HAS-BLED score, age, and CHA_2DS_2-VASc score should be interpreted cautiously.

Several limitations of the current trial should be acknowledged. First, the open-label design might have introduced reporting and performance bias. Placebo-controlled blinding was not feasible in this investigator-initiated trial because of manufacturing and distribution requirements and cost constraints. Although endpoints were adjudicated by committee members masked to treatment allocation, the same committee also served the data safety monitoring function; therefore, some risk of adjudication bias cannot be excluded. Second, nearly all PCI procedures in our trial were performed with intravascular imaging guidance—either optical coherence tomography or intravascular ultrasound—which is standard practice in Japan but remains uncommon in many other countries. This near-universal use of intravascular imaging guidance might limit the generalisability of our findings to settings in which PCI is guided mainly by angiography. Contemporary European and US guidelines recommend intravascular imaging in selected PCI settings, particularly for complex stenting and left main coronary interventions.[31,32] Trials comparing intravascular imaging guidance with angiography guidance for PCI have reported low rates of stent thrombosis in angiography-guided comparator groups, but the applicability of the OPTIMA-AF antithrombotic strategy to angiography-guided PCI remains uncertain..[33,34] Third, although the trial met the prespecified non-inferiority margin, the control event rate was lower than anticipated, so the fixed absolute margin represents a larger proportion of the observed risk than planned and could allow a clinically meaningful loss of efficacy. The upper bound of the 95% CI for the absolute risk difference was 3·6 percentage points, which is large relative to the observed incidence of the primary efficacy endpoint of 4·3% in the control group. For context, a margin corresponding to 50% of the observed 4·3% control-group incidence would be 2·15 percentage points, smaller than the upper bound of the observed 95% CI for the absolute risk difference. This post-hoc comparison underscores the need for cautious interpretation. In addition, the credibility of an active-controlled non-inferiority design depends on the control regimen representing contemporary usual best practice and being implemented as intended. The absolute efficacy of the 12-month dual-therapy control relative to DOAC monotherapy early after PCI remains uncertain, because a placebo-controlled comparison of antiplatelet continuation versus discontinuation is not feasible in this setting and has therefore not been evaluated in contemporary trials of PCI in patients with atrial fibrillation. Therefore, interpretation of non-inferiority relies on the clinical appropriateness of the active control relative to previous atrial fibrillation PCI trials and guideline-recommended practice, together with rigorous trial conduct (see appendix 2 [p 9] for the prespecified margin rationale). Accordingly, efficacy findings should be interpreted cautiously. Fourth, the study was conducted during the COVID-19 pandemic; recruitment was adversely affected and the enrolment period was extended. Follow-up procedures, including telephone contact when needed, were used to support endpoint ascertainment, but residual effects of the pandemic on trial conduct cannot be excluded. Fifth, the trial was underpowered to evaluate secondary endpoints including all-cause death and major bleeding, as well as infrequent ischaemic events such as stent thrombosis. Consequently, estimates for these outcomes—including the numerically higher incidence of all-cause death and some stent thrombosis outcomes—are imprecise and do not permit reliable inference, highlighting the need for adequately powered trials to evaluate efficacy and safety differences more robustly. Sixth, because race and ethnicity data were not collected, the ethnic composition of the trial population cannot be described precisely; however, recruitment exclusively in Japan suggests that the study population was likely to have been predominantly east Asian. Previous evidence suggests that east Asian and non-east Asian populations can differ in the balance of ischaemic and bleeding risks during antithrombotic therapy, which might limit the generalisability of our findings.[35–37] Finally, OPTIMA-AF included only 225 (21%) women, which might also have implications for the generalisability of the findings. In addition, because most patients had chronic coronary syndrome, caution is warranted in extrapolating our results to patients with acute coronary syndromes. Future studies are needed to establish whether the benefits of a short dual antithrombotic strategy can be extended to this high thrombotic risk subgroup.

In conclusion, among patients with atrial fibrillation and predominantly chronic coronary syndrome undergoing PCI with intravascular imaging guidance, 1-month dual antithrombotic therapy followed by DOAC monotherapy was non-inferior to 12-month therapy for death or thromboembolic events, and reduced major or clinically relevant non-major bleeding at 12 months, suggesting an overall favourable net clinical profile. Given the lower-than-anticipated event rates with a fixed absolute margin, efficacy should be interpreted with appropriate caution, including in light of numerically higher all-cause mortality in the 1-month group, which was largely reflected by numerically more non-cardiovascular deaths.

Contributors

YSo was the chief investigator and contributed to study design, data collection, data analysis, data interpretation, and manuscript writing. KKoz and YSa were co-principal investigators and contributed to study design, data interpretation, and manuscript writing. YH, GN, YM, KA, JA, KT, TM, and SH were steering committee members and contributed to study design, data interpretation, and manuscript writing. KKob, YI, TI, KH, SS, TW, TN, AK, RS, NO, YT, and AO were site principal investigators and contributed to data collection, data interpretation, and manuscript writing. KKa was the statistician and contributed to study design, data analysis, data interpretation, and manuscript writing. YSo, KKoz, YSa, and KKa accessed and verified the data. All authors had access to all data. All authors critically reviewed and revised the manuscript and approved the final version. All authors had final responsibility for the decision to submit for publication.

Declaration of interests

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Data sharing

De-identified individual participant data, the statistical code, and a data dictionary defining each field in the dataset will be made available until Dec 31, 2028, subject to approval by the Steering Committee. Access will be provided only to researchers with a methodologically sound research proposal, contingent upon approval by the trial steering committee and after signing a data access agreement. Requests should be directed to the corresponding author via email. The study protocol (version 1.38) and statistical analysis plan (version 4.0) are available in appendix 2.

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